

Nikhil Prakash

Research Interests

I'm interested in understanding the cognitive abilities of deep neural networks, such as reasoning and theory of mind, as well as exploring their downstream applications.

Education

- Sept 2022 – **Ph.D. & Masters**, *Northeastern University*, Boston, MA.
May 2027* Computer Science, Bau Lab (advisor: Prof. David Bau 🌐)
GPA: 4/4
- Aug 2016 – **Bachelor of Eng.**, *RV College of Eng.*, Bangalore, India
Aug 2020 Department of Telecommunication
Cumulative GPA: 8.31/10 (First Class with Distinction)

Peer-Reviewed Publications

Conference publications

- ArXiv 2026 **Agents of Chaos**
Natalie Shapira, Chris Wendler, Avery Yen, Gabriele Sarti, Koyena Pal, Olivia Floody, Adam Belfki, Alex Loftus, Aditya Ratan Jannali, Nikhil Prakash, et al.
- COLM 2026 **Pitfalls in Evaluating Interpretability Agents**
(submitted) Tal Haklay, Nikhil Prakash, Sana Pandey, Antonio Torralba, Aaron Mueller, Jacob Andreas, Tamar Rott Shaham, Yonatan Belinkov.
- ICML 2026 **The Dual Mechanisms of Spatial Reasoning in Vision–Language Models**
(submitted) Kelly Cui, Nikhil Prakash, Ayush Raina, David Bau, Antonio Torralba, Tamar Rott Shaham
- ICLR 2026 **Language Models use Lookbacks to Track Beliefs**
Nikhil Prakash, Natalie Shapira, Arnab Sen Sharma, Christoph Riedl, Yonatan Belinkov, Tamar Rott Shaham, David Bau, Atticus Geiger.

ICML 2025 **MIB: A Mechanistic Interpretability Benchmark**
Aaron Mueller, Atticus Geiger, Sarah Wiegrefe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fried Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, [Nikhil Prakash](#), Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, Martin Tutek, Amir Zur, David Bau, Yonatan Belinkov.

ICLR 2025 **NNsight and NDIF: Democratizing Access to Foundation Model Internals**
Jaden Fiotto-Kaufman, Alexander R Loftus, Eric Todd, Jannik Brinkmann, Caden Juang, Koyena Pal, Can Rager, Aaron Mueller, Samuel Marks, Arnab Sen Sharma, Francesca Lucchetti, Michael Ripa, Adam Belfki, [Nikhil Prakash](#), Sumeet Multani, Carla Brodley, Arjun Guha, Jonathan Bell, Byron Wallace, David Bau.

ICLR 2024 **Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking**
[Nikhil Prakash](#), Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, David Bau

IUI 2023 **Supporting Requesters in Writing Clear Crowdsourcing Task Descriptions Through Computational Flaw Assessment**
Zahra Nouri, [Nikhil Prakash](#), Ujwal Gadiraju, Henning Wachsmuth

[Journal publications](#)

Computational Linguistics **The Quest for the Right Mediator: A History, Survey, and Theoretical Grounding of Causal Interpretability**
Aaron Mueller, Jannik Brinkmann, Millicent Li, Samuel Marks, Koyena Pal, [Nikhil Prakash](#), Can Rager, Aruna Sankaranarayanan, Arnab Sen Sharma, Jiuding Sun, Eric Todd, David Bau, Yonatan Belinkov

[Workshop and demonstration publications](#)

ICLR 2026 **Constructive Circuit Amplification: Improving Math Reasoning in LLMs via Targeted Sub-Network Updates**
[Nikhil Prakash](#), Donghao Ren, Dominik Moritz, Yannick Assogba
Workshop on Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety across Modalities
Workshop on Representational Alignment

- NeurIPS 2025 **Language Models use Lookbacks to Track Beliefs**
Nikhil Prakash, Natalie Shapira, Arnab Sen Sharma, Christoph Riedl, Yonatan Belinkov, Tamar Rott Shaham, David Bau, Atticus Geiger
 First Workshop on CogInterp: Interpreting Cognition in Deep Learning Models
 Mechanistic Interpretability Workshop at NeurIPS 2025
- ICML 2023 **Discovering Variable Binding Circuitry with Desiderata**
 Xander Davies*, Max Nadeau*, Nikhil Prakash*, Tamar Rott Shaham, David Bau
 Workshop on Challenges in Deployable Generative AI.
- HCOMP 2021 **iClarify – A Tool to Help Requesters Iteratively Improve Task Descriptions in Crowdsourcing**
 Zahra Nouri, Nikhil Prakash, Ujwal Gadiraju, Henning Wachsmuth
 Work-in-Progress & Demonstration, Human Computation and Crowdsourcing 2021.
- NeurIPS 2020 **Conceptualization and Framework of Hybrid Intelligence Systems**
Nikhil Prakash and Kory W. Mathewson
 Human And Machine in-the-Loop Evaluation and Learning Strategies Workshop
- COMSNETS 2019 **A Grid-based Model for Generating Scale-Free Networks**
 Amit Kumar Verma and Nikhil Prakash
 Social Networking Workshop

Work Experience

- May–August 2026 **Goodfire AI, Research Fellow**
 Investigating the mechanisms responsible for variable binding in LMs.
- May–August 2025 **Apple Inc., Machine Learning Intern**
 Mechanistically investigating reasoning capabilities of language models, mentored by Yannick Assogba 🌐.
- July–August 2024 **Practical AI Alignment and Interpretability Research (Pr(Ai)²R) Group, Research Intern**
 Mechanistically investigating *theory of mind* capabilities in language models, mentored by Dr. Atticus Geiger 🌐.
- June–July 2023 **Stanford Existential Risks Initiative ML Alignment Theory Scholars (SERI-MATS)**
 Investigated superposition in attention heads of GPT2-XL that retrieve factual information from subject token residual stream, mentored by Neel Nanda 🌐.

- Jan–Aug 2022 **Max Planck Institute for Security and Privacy (MPI-SP), *Visiting Scholar***
 Worked under the supervision of Prof. Asia J. Biega 🌐 to conduct exploratory user studies for investigating well-being of microtask crowd-workers.
- May 2021–Sept 2022 **Delft University of Technology (TU Delft), *Research Assistance (Voluntary)***
 Worked under the supervision of Prof. Ujwal Gadiraju 🌐 and Prof. Henning Wachsmuth 🌐, to develop an assistive tool that would help crowdsourcing requesters in creating clear task descriptions quickly.
- July 2020–Jan 2022 **Accolite Software India, *Senior Software Engineer***
 Worked as full-stack engineer to design, develop, and maintain complex production-ready web applications. Primary technology stack was Angular, Java, and IBM DB2.
- Jan–May 2020 & June–Aug 2019 **Korea Advanced Institute of Science and Technology (KAIST), *Visiting Student Researcher***
 Worked under the supervision of Prof. Juho Kim 🌐 to develop an intelligent interactive system for personalizing food recipes based on users' preferences and dietary constraints.
- May–June 2019 **Samsung R&D Institute India - Bangalore, *Student Trainee***
 Worked as a student trainee at Samsung Research, Bangalore (SRIB) in the Service PF team to design and develop the offline newsfeed functionality in the Samsung Internet.
- June–July 2018 **Indian Institute of Technology, Ropar (IIT Ropar), *Research Intern***
 Worked under the supervision of Prof. Sudarshan Iyengar 🌐 to investigate the dynamics of collaboration process on English Wikipedia.

Invited Talks and Presentations

- 2026 Hima Lakkaraju's group at Harvard.
- 2025 Boston University NLP Group.
- 2025 Algorithms and Behavioral Science Coffee, MIT Economics.
- 2025 Apple AIML Visualization Team Meetings.
- 2025 Ploutos.dev.
- 2025 New England NLP 2025, Yale University.
- 2024 Practical AI Alignment and Interpretability Research Group.
- 2024 Computational Linguistics and Complex Social Networks Group at Indian Institute of Technology Gandhinagar.
- 2024 New England NLP 2024, Brown University.

Awards and Recognitions

- 2026 Silver Reviewer Award at ICML 2026.
- 2024 Top Reviewer at NeurIPS; Received complimentary registration, worth \$450.
- 2024 Google Gemma Academic Program GCP Credit Award, worth \$5k.
- 2022 Khoury College of Computer Sciences start-up fund, worth \$5k.
- 2020 NeurIPS complimentary registration (Travel Award), worth \$100.
- 2020 ACM CoDS-COMAD Student Travel Grant, worth \$80.

Voluntary Services

- 2026 Reviewer for ICML 2026, COLM 2026, NeurIPS 2026.
- 2025 Student Representative for Tenure Track Hiring Committee at Northeastern University.
- 2025 Volunteer for Code to PhD. Helping prospective PhD students with their applications.
- 2025 Reviewer for ICLR 2026.
- 2025 Reviewer for Mech Interp & CogInterp workshops at NeurIPS 2025.
- 2025 Reviewer for Visualization for AI Explainability workshop at IEEE VIS.
- 2025 Reviewer for BlackboxNLP at EMNLP 2025.
- 2025 Reviewer for XLLM-Reason-Plan workshop at COLM 2025.
- 2025 Reviewer for ICML 2025, COLM 2025, TMLR 2025, NeurIPS 2025 (main conference & workshop proposals).
- 2025 Reviewer for Actionable Interpretability Workshop at ICML 2025.
- 2024 Reviewer for ICLR 2025.
- 2024 Reviewer for Interpretable AI: Past, Present and Future, ATTRIB, and MINT workshops at NeurIPS 2024.
- 2024 Reviewer for NeurIPS 2024 (main conference & workshop proposals).
- 2024 Co-organizing Mechanistic Interpretability Social at ICLR 2024.
- 2023 Program committee member of ATTRIB 2023 workshop at NeurIPS.
- 2022 Student volunteer at Ph.D. admissions committee to review prospective Ph.D. student applications.
- 2019 Student volunteer at India HCI.

Engineering Skills

Languages	Python, C++, Java, JavaScript, HTML, CSS, SQL
Technologies & Frameworks	PyTorch, Transformers, NNSight, Transformer-lens, Scikit-learn, NLTK, Pandas, Numpy, Angular, Bootstrap, jQuery, Docker, Linux